

SelectKBest#

https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.SelectKBest.html

Description:

Select features according to the k highest scores. Read more in the User Guide. Function taking two arrays X and y, and returning a pair of arrays (scores, pvalues) or a single array with scores. Default is f_classif (see below “See Also”). The default function only works with classification tasks. Added in version 0.18.

Function Signature

```
class sklearn.feature_selection.SelectKBest(score_func=  
<function f_classif>, *, k=10)[source]#
```

Parameters:

score_funccallable, default=f_classif

kint or “all”, default=10

Attributes:

```
class sklearn.feature_selection.SelectKBest(score_func=  
<function f_classif>, *, k=10)[source]#
```

Parameters:

score_funccallable, default=f_classif

kint or “all”, default=10

Attributes:

```
fit(X, y=None)[source]#
```

Parameters:

Xarray-like of shape (n_samples, n_features)

yarray-like of shape (n_samples,) or None

Returns:

Parameters

scores_array

Scores of features.

pvalues_array

p-values of feature scores, None if score_func returned only scores.

Xarray

The training input samples.

yarray

The target values (class labels in classification, real numbers in regression). If the selector is unsupervised then y can be set to None.

Xarray

Input samples.

yarray

Target values (None for unsupervised transformations).

****fit_paramsdict**

Additional fit parameters.

input_featuresarray

Input features. If input_features is None, then feature_names_in_ is used as feature names in. If feature_names_in_ is not defined, then the following input feature names are generated: ["x0", "x1", ..., "x(n_features_in_ - 1)"]. If input_features is an array-like, then input_features must match feature_names_in_ if feature_names_in_ is defined.

paramsdict

Parameter names mapped to their values.

****paramsdict**

Estimator parameters.

Code Examples

```
>>> from sklearn.datasets import load_digits
>>> from sklearn.feature_selection import SelectKBest, chi2
>>> X, y = load_digits(return_X_y=True)
>>> X.shape
(1797, 64)
>>> X_new = SelectKBest(chi2, k=20).fit_transform(X, y)
>>> X_new.shape
(1797, 20)
```

Notes & Warnings

NOTE

See also `f_classif` ANOVA F-value between label/feature for classification tasks. `mutual_info_classif` Mutual information for a discrete target. `chi2` Chi-squared stats of non-negative features for classification tasks. `f_regression` F-value between label/feature for regression tasks. `mutual_info_regression` Mutual information for a continuous target. `SelectPercentile` Select features based on percentile of the highest scores. `SelectFpr` Select features based on a false positive rate test. `SelectFdr` Select features based on an estimated false discovery rate. `SelectFwe` Select features based on family-wise error rate. `GenericUnivariateSelect` Univariate feature selector with configurable mode.

Detailed Documentation

Documentation

Select features according to the k highest scores.

Read more in the User Guide.

Function taking two arrays X and y, and returning a pair of arrays (scores, pvalues) or a single array with scores. Default is `f_classif` (see below “See Also”). The default function only works with classification tasks.

Added in version 0.18.

Number of top features to select. The “all” option bypasses selection, for use in a parameter search.

p-values of feature scores, None if `score_func` returned only scores.

Number of features seen during fit.

Added in version 0.24.

Names of features seen during fit. Defined only when X has feature names that are all strings.

Added in version 1.0.

ANOVA F-value between label/feature for classification tasks.

Mutual information for a discrete target.

Chi-squared stats of non-negative features for classification tasks.

F-value between label/feature for regression tasks.

Mutual information for a continuous target.

Select features based on percentile of the highest scores.

Select features based on a false positive rate test.

Select features based on an estimated false discovery rate.

Select features based on family-wise error rate.

Univariate feature selector with configurable mode.

Ties between features with equal scores will be broken in an unspecified way.

This filter supports unsupervised feature selection that only requests X for computing the scores.

Run score function on (X, y) and get the appropriate features.

The training input samples.

The target values (class labels in classification, real numbers in regression). If the selector is unsupervised then y can be set to None.

Returns the instance itself.

Fit to data, then transform it.

Fits transformer to X and y with optional parameters fit_params and returns a transformed version of X.

Target values (None for unsupervised transformations).

Additional fit parameters.

Mask feature names according to selected features.

If `input_features` is `None`, then `feature_names_in_` is used as feature names in. If `feature_names_in_` is not defined, then the following input feature names are generated: `["x0", "x1", ..., "x(n_features_in_ - 1)"]`.

If `input_features` is an array-like, then `input_features` must match `feature_names_in_` if `feature_names_in_` is defined.

Transformed feature names.

Get metadata routing of this object.

Please check User Guide on how the routing mechanism works.

A `MetadataRequest` encapsulating routing information.

Get parameters for this estimator.

If `True`, will return the parameters for this estimator and contained subobjects that are estimators.

Parameter names mapped to their values.

Get a mask, or integer index, of the features selected.

If `True`, the return value will be an array of integers, rather than a boolean mask.

An index that selects the retained features from a feature vector. If `indices` is `False`, this is a boolean array of shape `[# input features]`, in which an element is `True` iff its corresponding feature is selected for retention. If `indices` is `True`, this is an integer array of shape `[# output features]` whose values are indices into the input feature vector.

Reverse the transformation operation.

X with columns of zeros inserted where features would have been removed by transform.

Set output container.

See [Introducing the set_output API](#) for an example on how to use the API.

Configure output of transform and fit_transform.

"default": Default output format of a transformer

"pandas": DataFrame output

"polars": Polars output

None: Transform configuration is unchanged

Added in version 1.4: "polars" option was added.

Set the parameters of this estimator.

The method works on simple estimators as well as on nested objects (such as Pipeline). The latter have parameters of the form <component>__<parameter> so that it's possible to update each component of a nested object.

Estimator parameters.

Reduce X to the selected features.

The input samples with only the selected features.

Selecting dimensionality reduction with Pipeline and GridSearchCV

Concatenating multiple feature extraction methods

Univariate Feature Selection

Release Highlights for scikit-learn 1.1

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